Rational approximation to $\log_2 3$

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Abstract

We prove that the irrationality exponent of $\theta = \log_2 3$ is at most three. More precisely, for every $\varepsilon > 0$, every sufficiently large positive integer q, and every integer p, one has $|\theta - p/q| \geq q^{-3-\varepsilon}$. The proof combines weighted interpolation at the points $(2^j, 3^j)$ with denominator clearing and a joint count of auxiliary and analytic Taylor indices. A curve inequality establishes the required interpolation, and a choice of rational parameters yields the bound. The proof is formalized in Lean.

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1 Introduction

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, log denotes the natural logarithm, and

$$\theta = \log_2 3 = \frac{\log 3}{\log 2}.$$

For an irrational real number ξ, its irrationality exponent $\mu(\xi)$ is the supremum of the real numbers ν for which

$$0 < \left|\xi - \frac{p}{q}\right| < q^{-\nu}$$

has infinitely many reduced solutions p/q with $q > 0$. Unique factorization shows that θ is irrational: a rational identity $\theta = p/q$, with $p, q > 0$, would imply $3^q = 2^p$. Dirichlet's approximation theorem therefore gives $\mu(\theta) \geq 2$. We prove the following upper bound.

Theorem 1.1. *For every $\varepsilon > 0$, there exists an integer $Q_\varepsilon \geq 2$ such that, for every $p \in \mathbb{Z}$ and every integer $q \geq Q_\varepsilon$,*

$$\left|\frac{\log 3}{\log 2} - \frac{p}{q}\right| \geq q^{-3-\varepsilon}. \tag{1.1}$$

Consequently,

$$2 \leq \mu(\theta) \leq 3.$$

Rational approximation to θ is equivalently a homogeneous linear-form problem:

$$\left|\theta - \frac{p}{q}\right| = \frac{|q \log 3 - p \log 2|}{q \log 2}.$$

For comparison, the linear-form estimate of Wu and Wang [9] implies $\mu(\theta) \leq 5.1163051$. The author's earlier preprint [1] gives the bound 5.104104, by a different construction with

symmetrized polynomials and exact finite certificates. Theorem 1.1 improves this numerical upper bound to three.

The geometric and analytic framework is inspired by OpenAI's proof of the exponent-two theorem for π [5], in particular its use of separated weights, interpolation, and determinants of common Taylor coefficients. Liu, Jiang, and Zhang [4] establish exponent two for every nonzero logarithm of a positive rational number. Related results for rational logarithms and compatible algebraic embeddings appear in [2]. For the quotient θ, we retain both base variables; primitive rational characters provide the curve rigidity needed for two-base interpolation. The final determinant estimate counts auxiliary and Taylor indices together.

1.1 Outline and order of choices

It suffices to exclude reduced approximations

$$|\theta - p/q| \le q^{-\nu} \tag{1.2}$$

with unbounded denominators for every rational $\nu > 3$. Assuming such approximations exist, we first fix rational seed parameters, then a finite auxiliary dimension and base weights. Only after these choices do we select finitely many exceptional fractions with sufficiently separated denominators. All these data remain fixed while the interpolation degree H tends to infinity.

Sections 2–3 establish the interpolation theorem from a local multiplicity comparison and a global curve inequality. Section 4 extracts a nonzero rational determinant and gives its arithmetic lower bound. Sections 5–6 obtain an incompatible upper bound by translating the rows to common analytic functions and counting distinct Taylor indices. Section 7 chooses the parameters and completes the proof. Section 8 describes the formal verification and its reproducible source package.

2 A local comparison

The varieties in this section and the next are over $\mathbb{C}$. We use the local multiplicity comparison underlying the separated-weight method of [5]; see also [4, Lemma 3].

A normal basis to a smooth subvariety means a collection of ambient vectors whose images form a basis modulo its tangent space. For an irreducible subvariety, all statements about these bases are at a general smooth point.

Lemma 2.1. *Let affine coordinates have positive rational degree weights W_a. Let commuting analytic vector fields D_b form a frame on an open set and have positive costs V_b. Suppose polynomials f_ℓ have weighted degree at most $N > 0$, and S is a codimension-k irreducible component, with $k \ge 1$, of their common zero set in that open set. If all further frame derivatives of every f_ℓ of cost less than ϵN vanish on S, then for a coordinate normal basis A and a frame normal basis B,*

$$\prod_{a\in A} W_a \le k!\epsilon^{-k} \prod_{b\in B} V_b \le k^k \epsilon^{-k} \prod_{b\in B} V_b. \tag{2.1}$$

Proof. Choose a smooth point of S with both basis properties and away from every other component. The commuting flows indexed by B, started on S, give analytic coordinates (z, b). Here z is on S and b is transverse. The derivative hypothesis says that each equation has an expansion

$$f_\ell(z,b) = \sum_{\sum_{j\in B} V_j \alpha_j \ge \epsilon N} f_{\ell,\alpha}(z) b^\alpha.$$

On the affine coordinate slice obtained by fixing coordinates outside A, the map to b is locally invertible. Substitution of its analytic functions $z = z(b)$ cannot decrease the weight of a surviving monomial. Consequently, the quotient by the restricted equations has length at least the number of nonnegative integer α with $\sum_{j\in B} V_j\alpha_j < \epsilon N$. The unit cubes with those lower corners cover the corresponding open real simplex, so this number is at least

$$\frac{(\epsilon N)^k}{k!\prod_{j\in B} V_j}.$$

The common zero on the coordinate slice is isolated. Generic k constant linear combinations of the restricted equations still have an isolated common zero there; their quotient length bounds the previous length from above. Weighted Bézout gives the upper bound $N^k/\prod_{a\in A} W_a$. One can obtain precisely this bound from ordinary isolated-zero Bézout: choose an integer M_0 with M_0W_a positive integers, and substitute $x_a = \lambda_a + z_a^{M_0W_a}$, choosing λ_a away from the coordinates of the point. The $\prod M_0W_a$ preimages are unramified, preserve local length, and the transformed equations have total degree at most M_0N. Comparison of the two length bounds proves the claim. □

3 Two-base interpolation

Fix positive integers m, K, positive rational w_0, v_0, v, σ with $v < 1$, and positive rational $w_1, \dots, w_m$. Index the two base coordinates first and the auxiliary coordinates in increasing order. Set

$$W = (w_0, w_0, w_1, \dots, w_m), \qquad V = (v_0, v_0, w_1/v, \dots, w_m/v), \qquad d = m + 2.$$

For an auxiliary multi-index α, write $w\alpha = \sum_i w_i\alpha_i$. Let $\mathcal{P}_W(H)$ be spanned by $Y^hZ^kX^\alpha$ with $w_0(h + k) + w\alpha \le H$. Let $\mathcal{J}_V(H)$ retain the formal coefficients $t^{\ell_1}s^{\ell_2}u^\beta$ for which $v_0(\ell_1 + \ell_2) + w\beta/v < H$.

Put $\epsilon = \sigma/(m + 3)$. Impose the separation inequalities

$$\prod_{a\in A} W_a > k^k\epsilon^{-k} \prod_{b\in B} V_b \tag{3.1}$$

for every two index sets of the same cardinality $1 \le k \le m+1$ whose largest differing auxiliary index lies in A. There are successive lower thresholds on the w_i ensuring all these inequalities. Indeed, at the largest differing auxiliary index the ratio contains the new weight linearly. Higher common indices cancel except for fixed factors $W_j/V_j = v$; the other weights are already fixed. These thresholds do not involve any rational slope.

For an explicit simultaneous choice put

$$L = \min(w_0, 1), \quad R_0 = \max(v_0, 1/v), \quad D = 1 + \sum_{k=0}^{d} k^k\epsilon^{-k}(R_0/L)^k,$$

using $0^0 = 1$. Write $x_0 = x_1 = 1$ and $x_{i+1} = w_i$ for $1 \le i \le m$. It suffices that all $x_i \ge 1$ and $x_i > D\prod_{j<i} x_j$ for every auxiliary index $i \ge 2$. Indeed $W_j \ge Lx_j$ and $V_j \le R_0x_j$. Cancel common indices from the two products. At the largest differing index, the growth condition exceeds the product of all earlier indices by the factor D; the remaining factors on its side are at least one. The definition of D then gives (3.1) for every cardinality simultaneously.

Proposition 3.1. *Suppose* (3.1) *and*

$$(1+3\sigma)^d K(w_0/v_0)^2 v^m < 1, \tag{3.2}$$

$$(1+\sigma)v < 1, \tag{3.3}$$

$$w_0/v_0 > 2(m+3)/\sigma. \tag{3.4}$$

Let $r_i = p_i/q_i$ be reduced positive fractions with

$$q_i v_0/w_0 > 1+\sigma. \tag{3.5}$$

At the centers $(2^j, 3^j, 0, \dots, 0)$, $0 \le j < K$, use the substitutions

$$Y = 2^j(1+t), \quad Z = 3^j(1+s), \quad X_i = \log(1+s) - r_i \log(1+t) + u_i. \tag{3.6}$$

Then $\mathcal{P}_W(H) \longrightarrow \bigoplus_{j<K} \mathcal{J}_V(H)$ is surjective for every sufficiently large H divisible by a fixed positive integer. Its degree threshold may depend on all the fixed data.

3.1 A curve inequality

On the complete smooth model of an irreducible affine curve define

$$D_W(C) = \sum_P \max\left(0, -\frac{\operatorname{ord}_P Y}{w_0}, -\frac{\operatorname{ord}_P Z}{w_0}, -\frac{\operatorname{ord}_P X_1}{w_1}, \dots, -\frac{\operatorname{ord}_P X_m}{w_m}\right).$$

An identically zero coordinate contributes no pole. At a branch through center j, put $t = Y/2^j - 1$, $s = Z/3^j - 1$, and $u_i = X_i - \log(1+s) + r_i \log(1+t)$, and set

$$h_P = \min\left(\frac{\operatorname{ord}_P t}{v_0}, \frac{\operatorname{ord}_P s}{v_0}, \frac{\operatorname{ord}_P u_i}{w_i/v} : 1 \le i \le m\right).$$

Orders of identically zero series are infinity. The formal coordinate change is invertible, so h_P is positive and finite. We prove

$$D_W(C) \ge (1+\sigma) \sum_{P \text{ centered}} h_P. \tag{3.7}$$

Suppose it fails. In particular the curve meets a center and neither Y nor Z vanishes identically. The leading dimensions of polynomials of degree at most N and of the conditions requiring vanishing below $(1+3\sigma)N$ are respectively

$$\frac{N^d}{d! \prod_a W_a}, \qquad \frac{K(1+3\sigma)^d N^d}{d! \prod_a V_a}.$$

Thus (3.2) supplies a nonzero such polynomial F_N for all sufficiently large N.

On $YZ \ne 0$ take the commuting frame

$$D_Y = Y\partial_Y - \sum_i r_i \partial_{X_i}, \quad D_Z = Z\partial_Z + \sum_i \partial_{X_i}, \quad D_i = \partial_{X_i}.$$

It preserves the degree bound and becomes $(1+t)\partial_t, (1+s)\partial_s, \partial_{u_i}$ in the jets. A frame word of cost at most σN applied to F_N has branch order at least $(1+2\sigma)Nh_P$. Its total poles on the curve are at most $ND_W(C)$. Failure of (3.7) would make the total zeros larger, unless that derivative vanished identically on the curve. Therefore every such word does vanish.

Consider the zero sets of frame derivatives of costs at most $r\epsilon N$, $0 \le r \le m+3$. Starting from the last, choose nested irreducible components containing the curve. Their dimensions

lie between one and $m+1$, so two adjacent ones coincide. Call the persistent component S. Further derivatives of the earlier equations of cost less than ϵN vanish on S. Lemma 2.1 applies to every pair of its normal bases.

If all auxiliary coordinate directions are tangent to S, its normal space has dimension $k = 1$ or 2. Coordinate and frame normal bases then use only base indices. The comparison gives $w_0/v_0 \le k/\epsilon \le 2/\epsilon$, contrary to (3.4).

Otherwise choose the largest nontangent auxiliary direction i. Some coordinate normal basis contains it. Higher auxiliary directions are zero in the normal quotient for both frames. Hence (3.1) forces every frame normal basis to contain i. The other frame vectors span the hyperplane $\ker(dX_i + r_i\,dY/Y - dZ/Z)$; their images fail to span the normal quotient. The tangent space of S is therefore in this hyperplane. On its function field we have

$$d(q_i X_i) = d\log(Z^{q_i}/Y^{p_i}). \tag{3.8}$$

Both rational functions in this identity are constant. To see this, if the character were nonconstant, choose a smooth point where it and X_i are regular and its differential is nonzero. Intersect S with affine hyperplanes through that point whose common tangent line is not in the kernel of this differential. The curve component through the point is smooth there, and the character remains nonconstant on it. Restrict (3.8) to this curve and take a complete smooth model. At one of its zeros or poles its logarithmic differential has nonzero residue, whereas an exact differential has residue zero. Once the character is constant, characteristic zero makes X_i constant too. Clearing denominators transfers these polynomial relations to the original curve. The center character values $(3^{q_i}/2^{p_i})^j$ are pairwise different, by unique factorization. The curve therefore meets at most one center.

If Y is nonconstant on that curve, choose integers a, b with $aq_i + bp_i = 1$. Writing $c = Z^{q_i}/Y^{p_i} \neq 0$ and $T = Y^a Z^b$, we have $Y = c^{-b}T^{q_i}$. At every branch through the possible center, T has the same nonzero residue T_j. The difference-of-powers factor has nonzero residue $q_i T_j^{q_i-1}$, whence

$$\operatorname{ord}_P(T - T_j) = \operatorname{ord}_P(Y - 2^j) \ge v_0 h_P.$$

The pole degree of Y is $q_i \deg(T)$. Zero–pole equality now implies

$$D_W(C) \ge \frac{q_i}{w_0}\deg(T) \ge \frac{q_i v_0}{w_0}\sum_P h_P > (1+\sigma)\sum_P h_P,$$

a contradiction. If Y is constant, $Z^{q_i} = cY^{p_i}$ makes Z constant. Some X_l is nonconstant, and the two logarithmic corrections vanish along the curve. Therefore

$$(w_l/v)\sum_P h_P \le \deg(X_l) \le w_l D_W(C),$$

contradicting (3.3). Curves missing all centers have contact sum zero. This proves (3.7) in every case.

3.2 Obtaining bounded-degree polynomials

Choose an integer R such that all $R/W_a, R/V_a$ are positive integers and $R \ge \max_a W_a$. Embed the affine space by its monomials of weighted degree at most R, and let X be the projective closure with $L = \mathcal{O}_X(1)$. The constant-coordinate chart is the original affine space. Pure powers and the constant monomial show on any normalized curve meeting this chart that

$$\deg L = R D_W(C). \tag{3.9}$$

Truncate both logarithms in auxiliary coordinate i with first omitted degree T_i such that $T_i v_0 > w_i/v$, obtaining regular coordinates t, s, u'_i. Every omitted term has order greater than $(w_i/v)h_P$, so the minimum defining h_P is unchanged. The coordinate change and its inverse preserve the entire weighted filtration. At each center form

$$I_j = (t^{R/v_0}, s^{R/v_0}, (u'_1)^{Rv/w_1}, \dots, (u'_m)^{Rv/w_m}).$$

Patch with the unit ideal elsewhere to get a coherent ideal I. Its branch order is Rh_P.

Let $p : X' \to X$ be the ordinary blow-up of I, and write $I\mathcal{O}_{X'} = \mathcal{O}_{X'}(-E)$, $A = p^*L$. On a noncontracted curve whose image meets the chart, (3.7) and (3.9) give nonnegative degree for $A-(1+\sigma)E$. Curves outside the chart miss E, and on contracted curves $-E$ has positive degree by relative ampleness. Thus $A-(1+\sigma)E$ is nef. For some integer $a > 1$, $aA - E$ is ample, and

$$A - E = \frac{a-1}{a(1+\sigma)-1}(A-(1+\sigma)E) + \frac{\sigma}{a(1+\sigma)-1}(aA-E)$$

is ample as well. These facts apply to the projective schemes in question; smoothness of X' is not needed; see [3, Corollary 1.4.10] and [6].

For large n, the graded-piece comparison and relative vanishing for the Rees algebra yield $p_*\mathcal{O}_{X'}(-nE) = I^n$ and $R^q p_*\mathcal{O}_{X'}(-nE) = 0$ for $q > 0$ [7, 8]. Projection, Leray and Serre vanishing give $H^1(X, I^n \otimes L^n) = 0$. Sections of L^n therefore realize all local classes modulo I^n. Each generator has formal weighted order at least R, so I^n is contained in the ideal of weight at least nR; the desired packet quotient is smaller. Completion does not change these finite-length quotients. Finally, Serre vanishing for the ideal sheaf of the projective embedding lifts every section of L^n, for large n, to a degree-n homogeneous polynomial in its monomial coordinates. On the affine chart this has weighted degree at most nR. Taking $H = nR$ proves Proposition 3.1.

4 A rational determinant and its lower bound

Fix a rational $\nu > 3$ and fractions satisfying (1.2), chosen large enough that $1 < r_i < 2$. Set $w_i = \lceil \log q_i \rceil$ and $w_* = \min_i w_i \geq 1$. Choose a rational $F > 2/v$, and use

$$T_i = \lceil F w_i / v_0 \rceil, \quad G_i(x) = \sum_{1 \leq n < T_i} \frac{(-1)^{n+1} x^n}{n}, \quad L_i = \operatorname{lcm}(1, \dots, T_i).$$

Since $T_i v_0 > w_i/v$, substituting G_i for both logarithms preserves the interpolation rank. The matrix, with all coefficient rows and all allowed monomial columns, is rational and has full row rank for large divisible H. Choose a nonzero square minor Δ_H using every row, and let its size be $M = M_H$.

Multiply column (h, k, α) by $\prod_i q_i^{\alpha_i}$, row $(j, \ell_1, \ell_2, \beta)$ by $\prod_i q_i^{-\beta_i}$, and every entry by $D_H = \prod_i L_i^{\lfloor H/w_i \rfloor}$. The resulting entries are integers: apart from integral factors they are coefficients of

$$2^{jh} 3^{jk} \binom{\alpha}{\beta} (1+t)^h (1+s)^k \prod_i [q_i G_i(s) - p_i G_i(t)]^{\alpha_i - \beta_i}.$$

Entries with some $\beta_i > \alpha_i$ are zero.

Let $\Lambda = 4 + \log 4$, and put $\bar{b}_H = (MH)^{-1} \sum_{\text{rows}} w\beta \leq v$. The elementary bound $\log \operatorname{lcm}(1, \dots, n) \leq \Lambda n$ suffices. For completeness, the ratio of the least common multiples through 2^k and 2^{k-1} divides $\binom{2^k}{2^{k-1}}$: each prime power in $(2^{k-1}, 2^k]$ contributes its underlying prime to that ratio and to the binomial coefficient. Summing the logarithmic bound over

dyadic intervals and enlarging to the next power of two gives the stronger constant $4\log 2$. The coarser constant Λ will suffice. Nonvanishing of the integral determinant gives

$$\frac{\log|\Delta_H|}{MH} \ge -1 + \bar{b}_H - E_{\mathrm{ar}}, \qquad E_{\mathrm{ar}} = \frac{\Lambda F m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{v}{w_*}. \tag{4.1}$$

Indeed the column logarithmic charge is at most MH, the replacement of $\log q_i$ by w_i in the row credit loses at most vMH/w_*, and $\log D_H/H$ is bounded by the first two terms in E_{ar}.

Weighted-simplex lattice counts give, with all data fixed,

$$M = \frac{K v^m H^{m+2}}{(m+2)! v_0^2 \prod_i w_i} + O(H^{m+1}). \tag{4.2}$$

This follows by rescaling the row simplex to the unit simplex, whose volume is $1/(m+2)!$.

5 Common analytic functions

For each auxiliary index a and column polynomial P, define

$$f_{a,P}(z,z') = [u^a]P(e^z, e^{z'}, (z' - \theta z + u_i)_{i=1}^m).$$

For a monomial column this is $\binom{\alpha}{a} e^{hz+kz'}(z'-\theta z)^{|\alpha|-|a|}$, or zero if $a \not\le \alpha$. At center j set $z = j\log 2 + \log(1+t)$, $z' = j\log 3 + \log(1+s)$. Writing $\delta_i = \theta - r_i$, let

$$B_i = G_i(s) - \log(1+s) - r_i(G_i(t) - \log(1+t)), \quad h_i = \delta_i \log(1+t) + B_i.$$

The exact identity $G_i(s) - r_i G_i(t) = z' - \theta z + h_i$ has no center-dependent error.

Lemma 5.1. *Every original row is a finite sum of scalar multiples of the common coefficient tests $[t^{\ell_1' } s^{\ell_2'}] f_{a,P}(z,z')$, with $a \ge \beta$, $wa \le H$, and $\ell_1' + \ell_2' < H/v_0$. The scalars are independent of P. Their number per row is at most*

$$C_H = (\lfloor H \rfloor + 1)^m (\lfloor H/v_0 \rfloor + 1)^2,$$

and each is bounded by $(\lfloor H \rfloor + 1)^m$ times

$$\exp\{-\nu w(a-\beta) + H E_{\mathrm{tr}}\}, \qquad E_{\mathrm{tr}} = \frac{\nu}{F} + \frac{\log 2}{v_0} + \frac{\nu + \log 12}{w_*}. \tag{5.1}$$

Proof. The scalar in the convolution with base indices d, e is

$$\binom{a}{\beta} [t^d s^e] \prod_i h_i^{a_i - \beta_i}, \qquad 0 \le d \le \ell_1, \quad 0 \le e \le \ell_2.$$

In its expansion let b_i factors be tails B_i. A nonzero contribution requires $\sum_i T_i b_i \le d + e < H/v_0$, hence $wb < H/F$. Because $w_i - 1 < \log q_i \le w_i$, its approximation factors are bounded by $\exp(-\nu w(a-\beta) + \nu H/F + \nu H/w_*)$. On $|t| = |s| = 1/2$, logarithms have modulus at most $\log 2 < 1$, and $|B_i| \le 3$. The two binomial factors cost at most $4^{H/w_*}$, the tails at most $3^{H/w_*}$, and Cauchy's coefficient estimate at most $2^{H/v_0}$. This proves the bound after summing at most $(\lfloor H \rfloor + 1)^m$ tail choices. Indeed, each auxiliary coordinate of a is at most $\lfloor H \rfloor$, since $w_i \ge 1$. For a fixed a, the convolution has exactly $(\ell_1 + 1)(\ell_2 + 1)$ possible splits, which is at most $(\lfloor H/v_0 \rfloor + 1)^2$. Thus the combined counting and scalar overhead per row is bounded by

$$Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)^2.$$

All data other than H are fixed, so $\log Q_H = O(\log H) = o(H)$. □

6 A joint Taylor-packing bound

We first record an elementary estimate that accounts simultaneously for auxiliary indices and analytic Taylor indices.

Lemma 6.1 (Packing with an extra coordinate)**.** *Let $c(x) = \sum_{i=1}^{r} c_i x_i$ on $\mathbb{N}^r$, where every $c_i > 0$, and let $S \subset \mathbb{N}^r$ have finite cardinality M. For $L \geq 0$, put*

$$N_c(L) = \#\{(n, x) \in \mathbb{N} \times \mathbb{N}^r : n + c(x) \leq L\}.$$

Then

$$\sum_{x \in S} c(x) \geq LM - N_c(L).$$

Proof. Discard the negative summands in $\sum_{x \in S}(L - c(x))$. For each remaining x, the integers $0 \leq n \leq L - c(x)$ give $\lfloor L - c(x) \rfloor + 1 \geq L - c(x)$ augmented points. These sets are disjoint for distinct x, and their union is contained in the set counted by $N_c(L)$. □

6.1 Uniform Taylor estimates

On the complex polydisc $|z|, |z'| \leq R_0 = 100K$, every common entire function $f_{a,P}$ associated to a monomial column has modulus at most

$$\mathcal{B}_H = \exp\left(H\left[\frac{100K}{w_0} + \frac{\log(600K)}{w_*}\right]\right).$$

Indeed, $1 < \theta < 2$, $h + k \leq H/w_0$, $|\alpha| \leq H/w_*$, and $\binom{\alpha}{a} \leq 2^{|\alpha|}$. Cauchy's estimate gives the same bound for its coefficient vectors in the normalized Taylor monomials $(z/R_0)^d (z'/R_0)^e$.

On the local contours $|t| = |s| = 1/2$, the translated variables satisfy $|z|, |z'| \leq R_0/2$. A coefficient test $[t^{\ell_1'} s^{\ell_2'}]$ of a normalized Taylor monomial is therefore bounded by

$$2^{\ell_1' + \ell_2'} 2^{-d-e}.$$

Expand a determinant of common coefficient tests row by row in these absolutely convergent Taylor series. If two rows use the same combined index (a, d, e), their coefficient vectors are proportional, and the corresponding determinant term vanishes. Thus every nonzero term has distinct combined indices.

6.2 The analytic upper bound

Assign to a combined index the cost

$$c(a, d, e) = \nu w a + \frac{d + e}{4},$$

and define

$$N(L) = \#\left\{(n, d, e, a) \in \mathbb{N}^{m+3} : n + \frac{d + e}{4} + \nu w a \leq L\right\}.$$

For any fixed $\lambda > 0$, Lemma 6.1 gives, in every nonzero Taylor term,

$$\nu \sum_{\text{rows}} w a + \frac{1}{4} \sum_{\text{rows}} (d + e) \geq \lambda H M - N(\lambda H). \tag{6.1}$$

Since $\log 2 \geq 1/2$,

$$2^{-d-e} \leq e^{-(d+e)/4} e^{-(d+e)/4}.$$

Combine the first exponential with the approximation cost in (6.1). The second exponential can be summed over all Taylor pairs, at a cost $(1-e^{-1/4})^{-2M}$ for the M rows. The coefficient determinant is bounded by $M!\mathcal{B}_H^M$. The local test orders contribute at most $2^{MH/v_0}$, and Lemma 5.1 gives a combined row-choice and scalar overhead of Q_H^M. It follows that the original nonzero determinant satisfies

$$\frac{\log|\Delta_H|}{MH} \le \nu\bar{b}_H - \lambda + \frac{N(\lambda H)}{MH} + E_{\mathrm{tr}} + E_{\mathrm{hol}} + \rho_H, \tag{6.2}$$

where

$$E_{\mathrm{hol}} = \frac{100K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(600K)}{w_*}, \tag{6.3}$$

$$\rho_H = \frac{\log M + \log Q_H - 2\log(1-e^{-1/4})}{H} \longrightarrow 0. \tag{6.4}$$

Here $M! \le M^M$, while $\log M$ and $\log Q_H$ are $O(\log H)$. The same geometric majorant justifies the infinite Taylor expansions and their determinant sums. Every estimate is uniform in the selected monomial columns and translated row choices.

6.3 Comparison with the arithmetic bound

The weighted-simplex count in dimension $m+3$ yields

$$N(\lambda H) \sim \frac{16\lambda^{m+3}H^{m+3}}{(m+3)!\nu^m\prod_i w_i}.$$

Together with (4.2), this gives

$$\frac{N(\lambda H)}{MH} \longrightarrow \frac{16v_0^2\lambda^{m+3}}{(m+3)K(\nu v)^m}. \tag{6.5}$$

Write

$$E = E_{\mathrm{ar}} + E_{\mathrm{tr}} + E_{\mathrm{hol}}.$$

Comparing (4.1) and (6.2), using $\nu > 1$ and the uniform bound $\bar{b}_H \le v$, and then letting H tend to infinity through its allowed multiples, gives the necessary inequality

$$\lambda - 1 - (\nu-1)v \le \frac{16v_0^2\lambda^{m+3}}{(m+3)K(\nu v)^m} + E. \tag{6.6}$$

Define $\lambda_* > 0$ by

$$\lambda_*^{m+2} = \frac{K(\nu v)^m}{16v_0^2}. \tag{6.7}$$

At this value the fraction in (6.6) equals $\lambda_*/(m+3)$. Thus

$$g_m := \frac{m+2}{m+3}\lambda_* - 1 - (\nu-1)v \le E. \tag{6.8}$$

We will choose the data so that the strict reverse inequality holds.

7 Explicit choices for every exponent above three

Fix a rational $\nu > 3$. Define rational numbers

$$e = \frac{\nu - 3}{2(2\nu - 3)}, \qquad s = 1 - e, \qquad v = s^3,$$
$$D = \nu s^2 - 1 - (\nu - 1)s^3, \qquad B = s - \frac{D}{4\nu s}, \qquad C = \frac{1}{2}\left(\frac{B^2}{v} + \frac{1}{B}\right).$$

Here $0 < s < 1$, and

$$D = (1 - s)((\nu - 1)s^2 - s - 1) > 0,$$
$$(\nu - 1)s^2 - s - 1 = \frac{\nu - 3}{2} + (\nu - 1)e^2 > 0.$$

Moreover $0 < D < \nu s^2$, so $0 < B < s$, and

$$G := \nu B^2 - 1 - (\nu - 1)v = \frac{D}{2} + \frac{D^2}{16\nu s^2} > 0. \tag{7.1}$$

It follows that

$$B^2 > v, \quad B^3 < v, \quad 1 < B^2/v < C < 1/B, \quad B^2 < Cv < 1, \quad CB < 1. \tag{7.2}$$

Choose a rational $F > 2/v$ large enough that $\nu/F < G/8$. For even integers m tending to infinity, set

$$J = \lfloor C^{m/2} \rfloor, \quad K = J^2, \quad w_0 = B^{-m}, \quad v_0 = 2Jv^{m/2}B^{-m}, \quad \sigma = \frac{1}{100(m+2)}. \tag{7.3}$$

All these weights are rational and $J \geq 1$. Exactly, $K(w_0/v_0)^2 v^m = 1/4$. In fact the inflated volume is less than $25/97 < 1$, since $(1 + 3/[100(m+2)])^{m+2} < 1/(1 - 3/100)$. Conditions (3.3) and (3.4) hold eventually: by (7.2),

$$K/w_0 \sim (CB)^m \longrightarrow 0,$$
$$v_0 \sim 2(\sqrt{Cv}/B)^m \longrightarrow \infty,$$
$$m/v_0 \longrightarrow 0,$$
$$w_0/v_0 \sim \tfrac{1}{2}(Cv)^{-m/2} \longrightarrow \infty \quad \text{exponentially.}$$

The last quantity eventually exceeds the quadratic expression $2(m+3)/\sigma$.

The value in (6.7) simplifies exactly to

$$\lambda_*^{m+2} = \frac{(\nu B^2)^m}{64}, \qquad \lambda_* \longrightarrow \nu B^2.$$

Thus $g_m \to G > 0$. Choose and fix an even m large enough for the geometric conditions and

$$g_m > G/2, \qquad E_0 := \frac{\Lambda F m + 2\log 2}{v_0} + \frac{100K}{w_0} < G/8. \tag{7.4}$$

Only now choose m fractions from the hypothetical unbounded sequence (1.2). Successively large denominators ensure $1 < r_i < 2$, (3.5), all separation inequalities, and

$$E_w := \Lambda \sum_i \frac{1}{w_i} + \frac{v + \nu + \log 12 + \log(600K)}{w_*} < G/8. \tag{7.5}$$

For the last condition it suffices first to require every w_i to exceed one fixed large constant, and then enforce successive separation. Unbounded denominators give unbounded $\lceil \log q \rceil$, even along the exceptional sequence.

Now all data are fixed. We have

$$E = \nu/F + E_0 + E_w < 3G/8 < g_m.$$

The interpolation proposition supplies the nonzero minors for all large divisible H. Taking H to infinity in their lower and upper bounds gives (6.8), contrary to $g_m > E$. This excludes (1.2) for each rational $\nu > 3$.

Given $\varepsilon > 0$, choose rational $3 < \nu < 3 + \varepsilon$. The set $\mathcal{E}$ of reduced fractions satisfying (1.2) is finite: its denominators are bounded, and the same inequality bounds its numerators. For $p/q \notin \mathcal{E}$, reduce the fraction to p'/q'. Then $q' \le q$, and

$$|\theta - p/q| = |\theta - p'/q'| > (q')^{-\nu} \ge q^{-\nu} \ge q^{-3-\varepsilon}.$$

If $\mathcal{E}$ is nonempty, then for its finitely many rational numbers, irrationality of θ gives a strictly positive minimum distance from θ. Since $q^{-3-\varepsilon}$ tends to zero, increasing the cutoff covers every representation of these fractions as well. If it is empty, no additional cutoff is needed. This proves (1.1). Together with Dirichlet's theorem it proves Theorem 1.1. □

8 Formal verification

The proof of Theorem 1.1 is formalized in Lean 4.34.1, using mathlib and the OpenAI formal library [5]. The approximation theorem and its irrationality-exponent corollary are in the module `LogRatioMain.lean`.

The formalization includes the local comparison, curve inequality, interpolation theorem, determinant estimates, and parameter construction. Its dependency closure was replayed in an initially empty Lean kernel environment, with only `propext`, `Classical.choice`, and `Quot.sound` permitted as axioms. The source supplement records the pinned dependencies, theorem-to-section correspondence, and commands for reproducing the verification.

AI Declaration

The author set the precise research direction and directed the investigation. OpenAI Codex (model `gpt-6-astra`) served as his research assistant in pursuing these directions, developing and testing mathematical arguments, implementing and debugging Lean proofs, preparing the manuscript, and identifying obstructions and unpromising approaches.

The author has personally verified the mathematical proofs and reviewed the accompanying Lean code. He takes full responsibility for the manuscript's accuracy, integrity, originality, and attribution.

Mathematical software and source availability

The manuscript and Lean sources, with pinned dependencies and reproduction instructions, are archived at doi:10.5281/zenodo.23256301 and maintained at `https://github.com/rohitkrjha/log23-measure`. The author's original Lean contributions are licensed under Apache 2.0; third-party sources retain their existing licenses. The preprint and its manuscript sources are licensed under Creative Commons Attribution 4.0 International.

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